
Public Transport Network Analysis

Graph Structure, Temporal Dynamics, and Accessibility

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This report examines the public transport networks of Canberra, Detroit, Paris, and Rome. It combines graph-theoretical measures, temporal traffic analysis, exact diameter computation, and earliest-arrival accessibility to describe how the networks are organized and how their services operate over time.

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Contents

1	Introduction	1
2	Construction of the Transport Graphs	1
3	Geographical Structure of the Networks	1
4	Question 1: Stops with the Largest Number of Departures	3
4.1	Canberra Results	3
5	Question 2: Nodes with the Highest Out-Degree	3
6	Question 3: Exact Diameter of the Main Component	4
6.1	Two-Sweep Initialization	4
6.2	iFUB Procedure	4
7	Question 4: Temporal Activity and Network Size	5
7.1	Temporal Profiles	5
7.2	Network Size and Execution Time	7
8	Question 5: Time-Dependent Accessibility	8
9	Computational Performance	8
10	Conclusions	9

1 Introduction

Public transport systems can be studied as networks in which stops are connected by scheduled trips. A purely static graph describes which stops are directly connected, whereas a temporal graph also represents when each trip departs and how long it takes. Both views are required to distinguish structural connectivity from the accessibility that passengers experience in practice.

The analysis covers four cities: **Canberra, Detroit, Paris, and Rome**. For every city, two temporal datasets are considered:

- a daily file, `network_temporal_day.csv`;
- a weekly file, `network_temporal_week.csv`.

The project addresses five questions. First, which stops generate the largest numbers of departures? Second, which stops have the greatest number of outgoing connections? Third, what is the exact diameter of the main connected component? Fourth, how does service volume evolve over time and how does computation time scale with network size? Fifth, how many stops can be reached from important origins within 30 minutes, one hour, and two hours?

2 Construction of the Transport Graphs

For each city, the node file contains stop identifiers, names, and geographical coordinates. The temporal connection file contains an origin stop, a destination stop, a departure timestamp, and an arrival timestamp. The duration of a connection is calculated as

$$\lambda = t_{\text{arrival}} - t_{\text{departure}}.$$

The graph-building function produces five objects for every city:

1. a directed graph $G = (V, E)$, where V is the set of stops and E is the set of direct connections;
2. a mapping from stop ID to stop name;
3. a static list of directed edges (u, v) ;
4. a mapping from stop ID to geographical coordinates (longitude, latitude);
5. a temporal edge list (u, v, t, λ) containing origin, destination, departure time, and duration.

These objects are stored in dictionaries indexed by city. This structure makes it possible to reuse the same data in all subsequent tasks without repeatedly reconstructing the networks.

3 Geographical Structure of the Networks

The geographical drawings reveal major differences in urban form. Canberra has a relatively dispersed structure, with a dense central area and several long peripheral branches. Paris presents a highly concentrated and interconnected central network. Detroit appears more grid-like, while Rome combines a dense irregular core with multiple radial corridors.

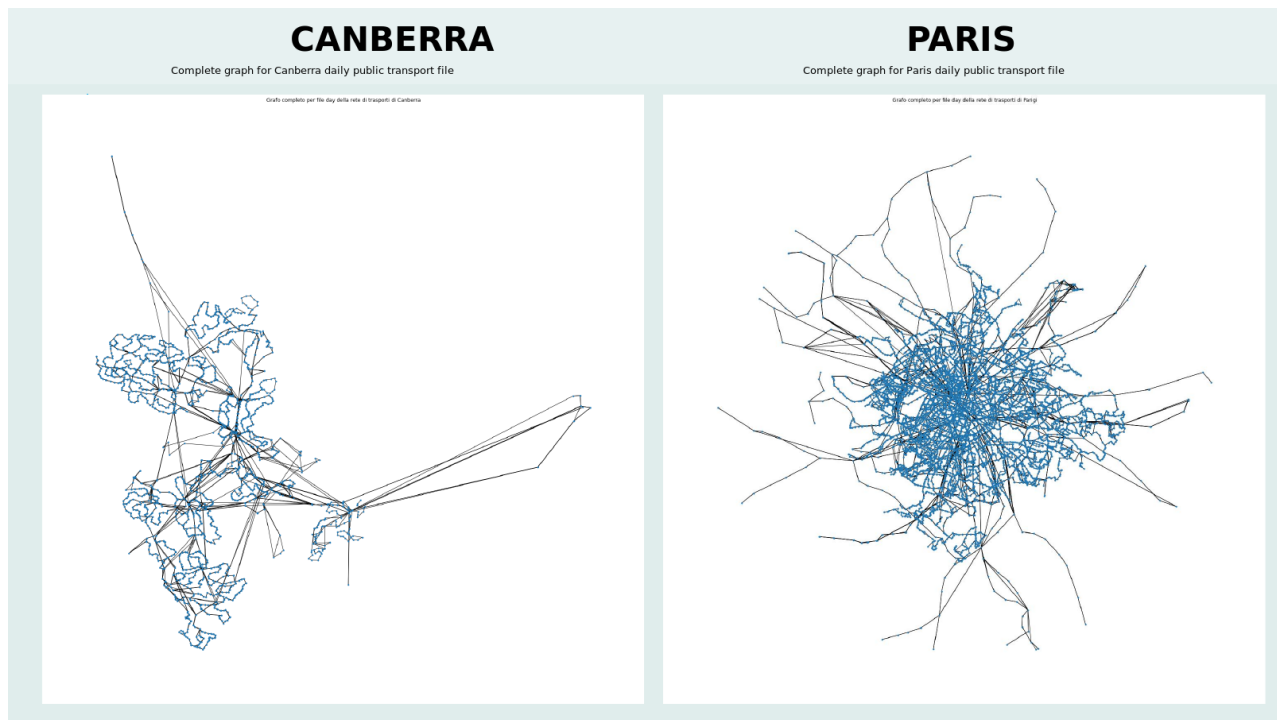


Figure 1: Geographical network structure of Canberra and Paris.

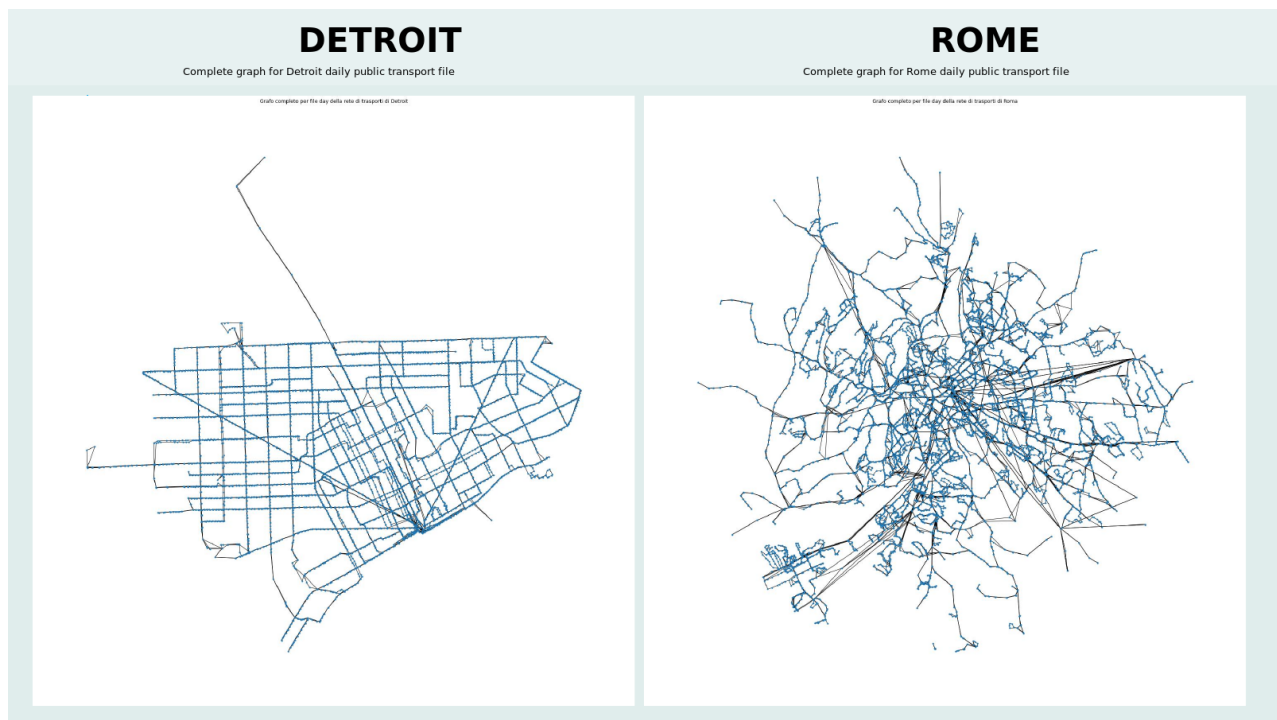


Figure 2: Geographical network structure of Detroit and Rome.

These visual differences help interpret later results. A geographically dense network may still have a large diameter if its connectivity is organized around long chains, while a dispersed network can have a smaller diameter when efficient cross-connections are present.

4 Question 1: Stops with the Largest Number of Departures

The first algorithm measures **service intensity**. Every temporal connection is counted at its origin stop. If several trips follow the same route, each trip contributes separately to the count. Therefore, the measure reflects how frequently services depart from a stop, not merely how many distinct destinations it serves.

For each city, the procedure is:

1. scan the temporal edge list and increment the counter of every origin stop;
2. convert the resulting dictionary into pairs (stop ID, departure count);
3. identify the ten largest counts by repeated maximum selection;
4. replace stop IDs with their names and store the ranking by city.

The complete edge list is scanned once. With a fixed top-10 selection, the overall complexity is approximately

$$O(C(n + m)),$$

where C is the number of cities, n the number of nodes, and m the number of temporal connections.

4.1 Canberra Results

Table 1: Canberra stops with the largest numbers of departures

Stop	Weekly	Daily	Week/day
Callum St after Hindmarsh Dr	2,963	524	5.654
Belconnen Community Station Arrivals	2,950	502	5.877
Westfield Bus Station Set Down Only	2,950	502	5.877
Westfield Bus Station	2,904	508	5.717
London Cct Legislative Assembly	2,845	508	5.599
Eastern Valley Way before Emu Bank Dr	2,785	480	5.802
London Cct Commonwealth Bank	2,632	474	5.553
Alinga St Health Building (Platform 13)	2,522	464	5.433
Launceston St opposite Phillip Oval	2,462	430	5.726
Eastern Valley Way after Emu Bank Dr	2,168	382	5.678

The weekly counts are consistently between about 5.4 and 5.9 times the daily counts. This indicates that the daily sample represents a relatively active service day rather than exactly one seventh of the weekly total.

5 Question 2: Nodes with the Highest Out-Degree

The second algorithm measures **structural connectivity**. The out-degree of a node is the number of directed graph edges leaving that node. Unlike departure frequency, repeated services on the same direct connection do not increase the out-degree.

The algorithm computes `G.out_degree()`, converts the result to a list of (node, degree) pairs, and repeatedly extracts the ten largest values. Since the number of selected nodes is fixed, the approximate complexity is $O(Cn)$.

Table 2: Canberra nodes with the highest out-degree

Stop	ID	Daily degree	Weekly degree
Civic Interchange Bay 6, Alinga St	14026	8	8
London Cct Legislative Assembly	1486	7	7
Morisset St After Crawford St	32495	7	7
Canberra Centre City	1249	6	6
Soward Way Homeworld	57	5	5
Comrie St opposite Erindale Centre Platform 2	161	5	7
Launceston St Phillip Oval	651	5	5
Brierly St Cooleman Court Platform 2	1103	5	5
Corinna St Westfield Woden	1160	5	–
Streeton Drive after Unwin Pl	1196	5	–
Soward Way after Anketell St	58	–	6
Commonwealth Av opposite Albert Hall	846	–	5

The distinction between Questions 1 and 2 is substantive. A stop can have many departures but only a few distinct outgoing links, or a moderate number of departures distributed across many destinations. Canberra’s highest observed out-degree is 8, recorded at Civic Interchange Bay 6 in both datasets.

6 Question 3: Exact Diameter of the Main Component

The diameter of a connected graph is the maximum shortest-path distance between any two nodes. It provides a topological measure of the network’s extent: a larger value means that some pair of stops is separated by a longer minimum sequence of connections.

The original graphs are directed. For this calculation, each graph is converted to an undirected graph and the largest connected component is selected. This prevents disconnected or isolated nodes from making the diameter undefined.

6.1 Two-Sweep Initialization

A highly connected node r is selected from the ranking produced in Question 2. A BFS from r identifies a distant node a . A second BFS from a identifies a distant node b . The shortest path between a and b provides an initial diameter estimate, and the central node of this path is used to initialize iFUB.

6.2 iFUB Procedure

The iterative Fringe Upper Bound algorithm begins with the eccentricity of the initial node u :

$$\text{ecc}(u) = \max_{v \in V} d(u, v).$$

It defines a lower bound $lb = \text{ecc}(u)$ and an upper bound $ub = 2\text{ecc}(u)$. A BFS groups nodes into distance layers. Starting from the most distant layer, the algorithm computes node eccentricities, updates the bounds, and stops when the lower bound is sufficient to establish the exact diameter. Each BFS costs $O(n + m)$, while the bound updates reduce the number of BFS executions compared with an exhaustive all-node calculation.

Table 3: Exact diameter results for the daily networks

City	Diameter	Main component	Starting stop (ID)
Canberra	51	2,520	Northbourne Av before Macarthur Av (2043)
Detroit	206	5,683	Rosa Parks & Spruce (5305)
Paris	159	10,644	La Boule Blanche (4767)
Rome	99	7,562	Cavalleggeri/S. Pietro (5992)

Detroit has the largest diameter, 206, despite having fewer nodes than Paris and Rome. This suggests a network with long topological chains and fewer shortcuts between some areas. Canberra has the smallest diameter, 51. Paris contains the largest main component but has a smaller diameter than Detroit, which is consistent with a denser set of alternative connections.

7 Question 4: Temporal Activity and Network Size

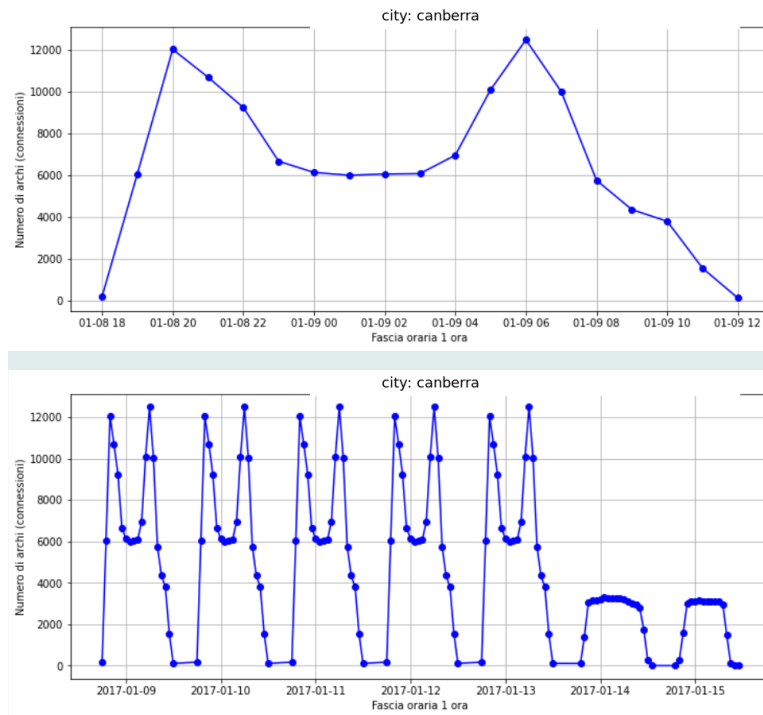
This analysis counts unique stops and temporal connections, aggregates departures by time interval, and records the computation time for every city. Departure timestamps are converted from Unix time to date-time values and rounded down to hourly intervals. The number of records within each interval forms the temporal traffic profile.

If r denotes the number of temporal records and k the number of time intervals, the approximate complexity is

$$O(C(n + r + k \log k)).$$

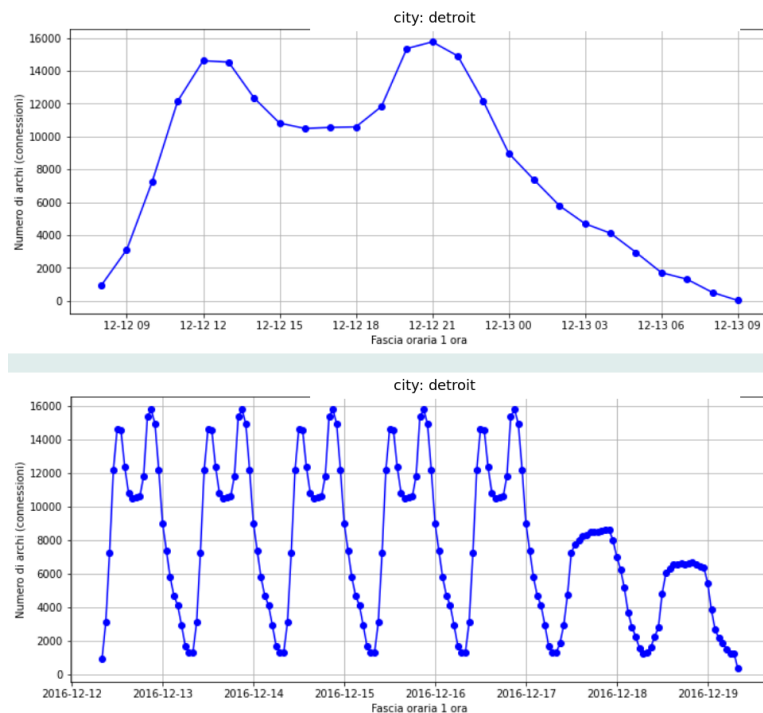
7.1 Temporal Profiles

The daily profiles show the development of service activity across a single observation period. The weekly profiles reveal repeated daily cycles, typically characterized by higher activity during daytime and a strong reduction overnight. The magnitude and shape of the cycles vary across cities, reflecting different network sizes and service schedules.



QUESTION 4: RESULTS FOR CITY CANBERRA

Figure 3: Complete hourly connection profiles for Canberra. The upper plot represents the daily file and the lower plot the weekly file.



QUESTION 4: RESULTS FOR CITY DETROIT

Figure 4: Complete hourly connection profiles for Detroit. The upper plot represents the daily file and the lower plot the weekly file.

7.2 Network Size and Execution Time

Table 4: Network size and execution time

City	Nodes	Connections		Time (s)		Edge ratio
	Daily	Daily	Weekly	Daily	Weekly	Week/day
Canberra	2,764	124,305	703,614	0.1842	0.5032	5.660
Detroit	5,683	214,863	1,317,208	0.2199	0.6328	6.132
Paris	11,950	1,823,872	11,507,679	0.9759	4.2837	6.311
Rome	7,869	1,051,211	6,828,376	0.6747	2.7165	6.497
Mean ratio						6.150

Paris is the largest network in terms of both nodes and temporal connections. Rome ranks second by number of connections, followed by Detroit and Canberra. Weekly files contain approximately six times as many connections as daily files. However, execution time does not increase at exactly the same rate: the mean weekly-to-daily runtime ratio for this task is about 3.51.

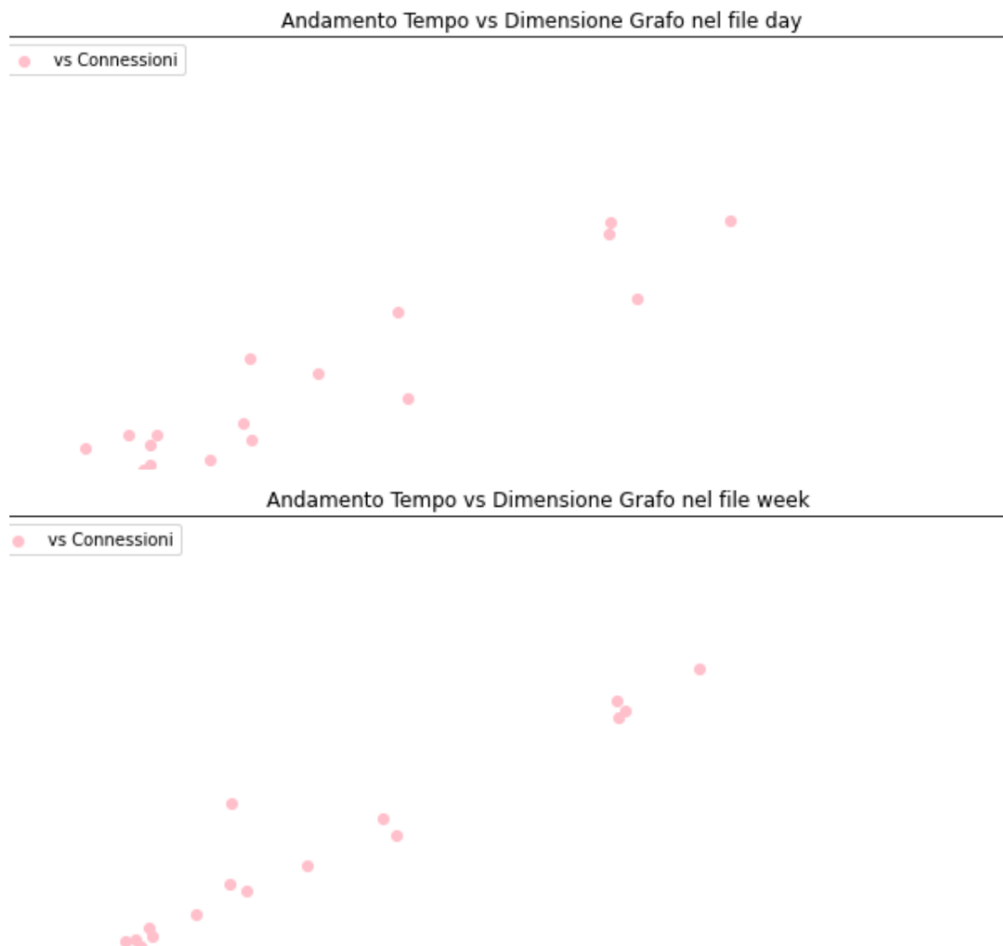


Figure 5: Relationship between the number of connections and execution time for the daily file (upper panel) and weekly file (lower panel).

The scatter plots show a clear positive association between graph size and computation time. Paris and Rome occupy the upper-right part of both plots because they combine large numbers of connections with longer processing times.

8 Question 5: Time-Dependent Accessibility

The final analysis evaluates how many stops are reachable from each of the ten high-out-degree nodes within three thresholds: 30 minutes, one hour, and two hours. Unlike static shortest paths, this problem must respect departure times, waiting times, and chronological feasibility.

For an origin node x , the earliest available departure defines t_α . The maximum observation horizon is

$$t_\omega = t_\alpha + 7,200 \text{ seconds.}$$

The earliest-arrival algorithm initially assigns infinity to every stop and t_α to the origin. Temporal edges are sorted by departure time. A connection (u, v, t, λ) is feasible if

$$t \geq T(u) \quad \text{and} \quad t + \lambda \leq t_\omega,$$

where $T(u)$ is the earliest known arrival time at u . If $t + \lambda < T(v)$, the arrival time of v is updated. Finally, the algorithm counts the stops satisfying each time threshold.

Table 5: Time-dependent accessibility from leading Canberra nodes, daily file

Origin stop (ID)	30 min	1 hour	2 hours
Civic Interchange Bay 6, Alinga St (14026)	13	39	115
London Cct Legislative Assembly (1486)	22	325	1,749
Morisset St After Crawford St (32495)	45	75	87
Canberra Centre City (1249)	103	474	2,154
Soward Way Homeworld (57)	186	519	2,192
Comrie St opposite Erindale Centre Platform 2 (161)	75	462	2,163
Launceston St Phillip Oval (651)	85	336	1,619
Brierly St Cooleman Court Platform 2 (1103)	100	535	2,195
Corinna St Westfield Woden (1160)	135	412	2,214
Streeton Drive after Unwin Pl (1196)	57	356	1,857

Accessibility increases sharply with the time horizon, but the growth is not uniform across origins. Morisset St, for example, reaches 45 stops in 30 minutes but only 87 in two hours. By contrast, Soward Way Homeworld reaches 186 stops in 30 minutes and 2,192 in two hours. This confirms that out-degree alone does not determine temporal accessibility: schedule alignment, waiting time, and downstream connections also play central roles.

9 Computational Performance

Table 6: Execution times of the principal operations

Operation	Daily file	Weekly file	Week/day ratio
File reading	5 min 58.2212 s	44 min 47.6525 s	7.5027
Question 1: departure counts	18.1220 s	2 min 58.6406 s	9.8570
Question 2: out-degree	0.1439 s	0.9925 s	6.8980
Question 3: diameter	16.2017 s	23.1551 s	1.4290
Question 5: accessibility	2 min 51.8669 s	12 min 16.2981 s	4.2850
Mean ratio			5.9943

File reading is a major cost, especially for the weekly datasets. Question 2 is the fastest operation because out-degree is directly available from the static graph. Question 5 is computationally demanding because each origin requires the temporal edge sequence to be processed under chronological constraints. Question 3 scales particularly well: its weekly runtime is only 1.429 times the daily runtime, showing the effectiveness of the iFUB bounds.

10 Conclusions

The project provides complementary views of public transport performance. Departure frequency identifies operational hubs, while out-degree identifies structurally well-connected stops. These measures are related but not interchangeable. The diameter analysis describes the maximum topological separation inside the main network component, and its results show that a larger node count does not necessarily imply a larger diameter.

Temporal aggregation reveals clear service cycles and demonstrates the greater computational burden of weekly datasets. The earliest-arrival analysis adds the passenger's perspective: a route is useful only if its timing is compatible with previous arrivals and the chosen travel horizon.

Overall result. The combination of directed graphs, temporal edges, BFS-based diameter estimation, and earliest-arrival calculations gives a detailed picture of network activity, structural connectivity, geographical organization, computational scale, and practical accessibility across the four cities.